

Lesson 1: System Administration and Maintenance (Linux)

Abbreviations

- **OS:** Operating System
- **PC:** Personal Computer
- **GNU:** GNU's Not Unix
- **IEEE:** Institute of Electrical and Electronics Engineers
- **FSF:** Free Software Foundation
- **LAMP:** Linux, Apache, MySQL, Perl/PHP/Python
- **GPL:** General Public License
- **GUI:** Graphical User Interface
- **CLI:** Command Line Interface
- **MS:** Microsoft
- **I/O:** Input/Output
- **BSD:** Berkeley Software Distribution
- **SCSI:** Small Computer System Interface
- **NCP:** NetWare Core Protocol
- **SMB:** Server Message Block
- **LAN:** Local Area Network
- **NFS:** Network File System
- **Btrfs:** B-tree File System
- **VM:** Virtual Machine
- **NUMA:** Non-Uniform Memory Access
- **ZSTD:** Zstandard
- **RAM:** Random Access Memory
- **initramfs:** Initial RAM File System
- **USB:** Universal Serial Bus
- **IBM:** International Business Machines
- **BIOS:** Basic Input/Output System
- **AMD:** Advanced Micro Devices
- **GPU:** Graphics Processing Unit
- **HMM:** Heterogeneous Memory Management
- **SVM:** Shared Virtual Memory
- **ACPI:** Advanced Configuration and Power Interface
- **VMM:** Virtual Machine Monitor
- **CPU:** Central Processing Unit
- **IPC:** Inter-Process Communication
- **NT:** New Technology
- **IoT:** Internet of Things
- **SUS:** Single UNIX Specification
- **POSIX:** Portable Operating System Interface
- **ALSA:** Advanced Linux Sound Architecture

- **SELinux:** Security-Enhanced Linux
- **sh:** Bourne Shell
- **bash:** GNU Bourne-Again Shell
- **csch:** C Shell
- **ksh:** Korn Shell
- **fish / fsh:** Friendly Interactive Shell
- **zsh:** Z Shell
- **tcsh:** TENEX C Shell

Enumerations

- **Operating Systems Comparison (Pros and Cons):**
 - *Windows Pros:*
 - Support for all hardware
 - Ease of use
 - Software support
 - Plug and play feature
 - Desktop and touchscreen.
 - *Windows Cons:*
 - Virus attacks
 - Paid software
 - High computer resources
 - Technical support.
 - *Mac Pros:*
 - Fewer virus attacks
 - Good customer support
 - Similar GUI for all products
 - Performance and long life
 - Default apps.
 - *Mac Cons:*
 - Expensive
 - Fewer games and software
 - No hardware customization.
 - *Linux Pros:*
 - Open-source
 - No need for anti-virus
 - Powerful CLI
 - Low system specifications
 - Good at multitasking
 - Less disk space needed.
 - *Linux Cons:*
 - Hardware drivers
 - Learning curve

- Software alternative
- Games.

- **Fundamental Differences (Linux vs. Windows):**

1. Linux is yours, Windows is still theirs.
2. Linux needs root privileges to be able to execute codes.
3. Linux does not need any anti-virus software (No OS is totally secure).
4. Windows users are encouraged to pay for software that is only about 20% effective and severely limits performance.
5. Linux uses a centralized software approach, Windows does not.
6. Most new-user-friendly Linux distributions come with the most needed software already installed.
7. Linux software is installed in predetermined directories; it cannot be moved or installed elsewhere.
8. Proprietary drivers are available through online repositories; Windows drivers include unwanted software or adware.
9. Linux updates everything at once and never restarts without permission; Windows updates select software and restarts without consent.
10. Linux uses /dev for drives; Windows uses alphabet letters.

- **Mistakes from New Linux Users:**

1. Choose wrong hardware.
2. Choose wrong Linux distributions.
3. Linux is not Windows (expectation).
4. User uses Google to find software.
5. User doesn't take time to learn Linux.

- **Kernel Objectives:**

- To establish communication between user-level applications and hardware.
- To decide the state of incoming processes.
- To control disk management.
- To control memory management.
- To control task management.

- **What Kernel Manages:**

- CPU/GPU
- Memory
- Input/Output (I/O) devices
- Resource management
- Memory management
- Device management
- System calls
- Kernel Security & Protection

- **Kernel Functions:**

- Process Management
- Memory Management
- Device Management
- File System Management
- Resource Management
- Security and Access Control
- Inter-Process Communication (IPC)
- Task Scheduling, Load Balancing, Error Handling

- **Types of Kernels:**

1. Monolithic Kernel
2. Microkernel
3. Hybrid Kernel
4. Exo Kernel
5. Nano Kernel
6. Virtual Machine (VM) Kernel or Hypervisor
7. Embedded Kernel

- **Different Types of Shells in Linux:**

1. Bourne Shell (sh)
2. GNU Bourne-Again Shell (bash)
3. C Shell (csh)
4. Korn Shell (ksh)
5. Friendly Interactive Shell (fish / fsh)
6. Z Shell (zsh)
7. TENEX C Shell (tcsh)

- **Shell Prompts by Type:**

- *Bourne shell (sh)*: Root: # | Non-root: \$
- *GNU Bourne-Again shell (bash)*: Root: bash-VersionNumber# | Non-root: bash-VersionNumber\$
- *C shell (csh)*: Root: # | Non-root: %
- *Korn shell (ksh)*: Root: # | Non-root: \$
- *Z Shell (zsh)*: Root: hostname># | Non-root: hostname>\$

Lesson 2: File System

Abbreviations

- **FS:** File System
- **TB:** Terabyte
- **EB:** Exabyte
- **COW:** Copy-on-Write
- **NAS:** Network Attached Storage
- **XFS:** X File System
- **ZFS:** Zettabyte File System
- **FAT:** File Allocation Table
- **NTFS:** New Technology File System
- **exFAT:** Extended File Allocation Table
- **ReFS:** Resilient File System
- **CD / CD-ROM:** Compact Disc / Compact Disc Read-Only Memory
- **HDD:** Hard Disk Drive
- **FHS:** Filesystem Hierarchy Standard
- **FTP:** File Transfer Protocol
- **USR:** User System Resources

Enumerations

- **Linux File Systems:**
 - Minix file system
 - Extended file system family (ext, ext2, ext3, ext4)
 - Btrfs
 - ReiserFS
 - XFS
 - ZFS
 - JFS
- **Common Use Cases for ZFS:**
 - Enterprise Storage Solutions
 - Backup and Disaster Recovery
 - File Server and NAS
- **Components of NFS:**
 - NFS Server
 - NFS Client
- **MS Windows File Systems:**
 - FAT
 - NTFS
 - exFAT
 - Live File System
 - Resilient File System (ReFS)

- **Linux Directory Structure:**

- / (Primary root directory)
- /bin (Essential command binaries)
- /boot (Boot loader files)
- /dev (Device files)
- /etc (Host-specific system-wide configuration files)
- /home (User home directories)
- /lib (Essential shared libraries and kernel modules)
- /media (Mount points for removable media)
- /mnt (Temporarily mounted filesystems)
- /opt (Add-on application software packages)
- /proc (Virtual filesystem for process and kernel information)
- /root (Home directory for the root user)
- /run (Run-time variable data)
- /sbin (Essential system administrative binaries)
- /srv (Site-specific data served by the system)
- /sys (Device, driver, and kernel feature information)
- /tmp (Temporary files)
- /usr (Secondary hierarchy for user data and applications)
- /var (Variable data files)

Lesson 3: Linux Distributions

Abbreviations

- **YaST:** Yet another Setup Tool
- **DVD:** Digital Versatile Disc

- **RHEL:** Red Hat Enterprise Linux
- **KVM:** Kernel-based Virtual Machine

Enumerations

- **Linux Distributions Listed:**

1. Slackware (July 1993)
2. Debian (September 1993)
3. SUSE (1994)
4. Red Hat (May 1995)
5. Knoppix (September 2000)
6. Gentoo (March 2002)
7. Arch Linux (March 2002)
8. Fedora (November 2003)
9. CentOS (2004)
10. Ubuntu (October 2004)
11. Alpine Linux (August 2005)
12. Oracle Linux (October 2006)
13. Musix (December 2006)
14. Android (September 2008)
15. Mageia (June 2011)

Lesson 4: Desktop Environments, Package Management, and Boot Loaders

Abbreviations

- **DE:** Desktop Environment
- **SSD:** Solid-State Drive

- **MBR:** Master Boot Record
- **GPT:** GUID Partition Table
- **GUID:** Globally Unique Identifier
- **LightDM:** Lightweight Display Manager
- **SDDM:** Simple Desktop Display Manager
- **GDM / GDM3:** GNOME Display Manager
- **LXDM:** LXDE Display Manager
- **XDM:** X Display Manager
- **Ly:** Lightweight
- **KDM:** KDE Display Manager
- **WDM:** WINGS Display Manager
- **MDM:** MDM Display Manager
- **CDM:** Console Display Manager
- **SLiM:** Simple Login Manager
- **XDMCP:** X Display Manager Control Protocol
- **KDE:** K Desktop Environment
- **GNOME:** GNU Network Object Model Environment
- **LXDE:** Lightweight X11 Desktop Environment
- **LXQt:** Lightweight Qt Desktop Environment
- **XFCE / Xfce:** XForms Common Environment
- **BSPWM:** Binary Space Partitioning Window Manager
- **CDE:** Common Desktop Environment
- **UKUI:** Ubuntu Kylin User Interface
- **KIS:** Keep It Simple
- **ICCCM:** Inter-Client Communication Conventions Manual
- **EWMH:** Extended Window Manager Hints
- **APK:** Alpine Package Keeper / Android Package Kit
- **MSI:** Microsoft Installer
- **RPM:** RPM Package Manager (originally Red Hat Package Manager)
- **YUM:** Yellowdog Updater, Modified
- **DNF:** Dandified YUM
- **APT:** Advanced Package Tool
- **HTTP:** Hypertext Transfer Protocol
- **d-i:** Debian-Installer
- **Wubi:** Windows-based Ubuntu Installer
- **GRUB:** GRand Unified Bootloader
- **LILO:** Linux Loader
- **UEFI:** Unified Extensible Firmware Interface
- **EFI:** Extensible Firmware Interface
- **NTLDR:** NT Loader
- **BOOTMGR:** Windows Boot Manager
- **BURG:** Brand-new Universal loader from GRUB
- **RAID:** Redundant Array of Independent Disks
- **GHz:** Gigahertz

- **GB:** Gigabyte

Enumerations

- **Major Display Renders:**

- Xorg / X Window System (X11)
- Wayland

- **Common Linux Partitioning Utilities:**

1. fdisk
2. parted
3. gdisk
4. cfdisk
5. gparted
6. lsblk

- **Linux Display Managers:**

1. GNOME Display Manager (GDM/GDM3)
2. Simple Desktop Display Manager (SDDM)
3. Lightweight Display Manager (LightDM)
4. LXDE display manager (LXDM)
5. X Display Manager (XDM)
6. Lightweight (Ly)
7. KDE Display Manager (KDM)
8. WINGS Display Manager (WDM)
9. MDM Display Manager (MDM)
10. Console Display Manager (CDM)
11. SLiM (Simple Login Manager)
12. Entrance
13. XDMCP (X Display Manager Control Protocol)
14. emptyty
15. greetd

- **Linux Desktop Environments and Window Managers:**

- KDE Plasma
- MATE
- GNOME
- Cinnamon
- Budgie
- LXQt
- LXDE
- XFCE
- Deepin
- Unity

- Pantheon
- Lumina
- Enlightenment
- i3
- Ambient
- BSPWM
- CDE
- Trinity
- UKUI
- Sugar
- Openbox
- Fluxbox
- Awesome
- Hyprland

- **Steps in Linux Installation:**

- Keyboard settings
- Partition
- Set Timezone
- User Information
- Install files (copy files to HDD)

- **Package Formats and Consuming Systems:**

- APK (Alpine) – Alpine Linux
- APK (Android) – Android
- AppImage – Linux distribution-agnostic
- Deb – Debian, Ubuntu, Xubuntu, Linux Mint
- ebuild – Gentoo Linux
- eopkg – Solus
- Flatpak – Linux distribution-agnostic
- PKG – OS X, iOS, PlayStation 3, Solaris, SunOS, UNIX System V, Symbian, BeOS, Apple Newton
- .pkg.tar.zst – Arch Linux
- PUP and PET – Puppy Linux
- RPM – Red Hat Enterprise Linux, Fedora, CentOS, SUSE Linux Enterprise Server, openSUSE
- Snap – Linux distribution-agnostic (mainly Ubuntu)
- MSI – Microsoft Windows

- **Popular Package Managers:**

- npm – Node.js
- pip – Python
- vcpkg – C and C++
- Homebrew – macOS

- apt – Debian packages
- yum and dnf – Fedora and Red Hat Enterprise Linux
- pacman – Arch Linux
- zypper – openSUSE / SUSE
- nala – APT frontend

- **RPM Modes:**

- Query, Verify, Install, Upgrade, Erase, Show querytags, Show configuration

- **Linux System Installers:**

1. Anaconda
2. Debian-Installer
3. DrakX
4. Preseed
5. Snap
6. Wubi
7. Calamares

- **Tasks Performed by Debian-Installer:**

- Detects hardware
- Detects installation media
- Configures the network
- Configures APT / selects mirrors
- Sets up user accounts
- Detects other operating systems

- **Boot Loaders:**

1. GNU GRUB (Grand Unified Bootloader)
2. LILO (Linux Loader)
3. rEFInd
4. BootX (Apple / Linux)
5. NTLDR
6. BURG
7. SYSLINUX

- **System Requirements by Distribution:**

- *Arch Linux*: 2GB RAM (4GB+ rec.), 10GB HDD, 64-bit CPU Dual Core 2GHz+.
- *Kali Linux*: 2GB RAM (8GB+ rec.), 20GB HDD, 32/64-bit CPU Dual Core 2GHz+.
- *Linux Mint*: 4GB RAM+, 20GB HDD (100GB rec.), 64-bit CPU Single Core 2GHz+.
- *Ubuntu Linux*: 4GB RAM (8GB+ rec.), 40GB HDD, 64-bit CPU Quad Core 2.5GHz+.
- *Fedora Linux*: 8GB RAM (12GB+ rec.), 40GB HDD, 64-bit CPU Quad Core 2GHz+.
- *SUSE Linux*: 4GB RAM (8GB+ rec.), 30GB HDD, 64-bit CPU Quad Core 2GHz+.

Lesson 5: Basic Commands

Abbreviations

- **UID:** User Identification / User ID
- **GID:** Group Identification / Group ID
- **PAM:** Pluggable Authentication Modules
- **ASCII:** American Standard Code for Information Interchange
- **IP:** Internet Protocol
- **SSH:** Secure Shell
- **DNS:** Domain Name System
- **PING:** Packet Internet Groper

Enumerations

- **Types of Linux Paths:**

1. Relative paths
2. Absolute paths

- **Linux Directory Commands:**

- pwd (print working directory)
- mkdir (make directory)
- rmdir (remove directory)
- ls (list directory contents)
- cd (change directory)

- **Linux File Commands:**

- touch (create empty files)
- cat (concatenate and display file content)
- echo (output a string)
- printf (formatted printing)
- rm (remove files or directories)
- cp (copy files or directories)
- mv (move files or directories)
- rename (rename multiple files)
- head (display first lines)
- tail (display last lines)
- tac (display file contents in reverse order)
- more (display file screen-by-screen)
- less (enhanced file viewing)

- **Navigation Keys in more Command:**

- ENTER: Scroll down page by line
- Space bar: Move to the next page
- b key: Move to the previous page
- / key: Search the string

- **Linux User & Group Commands:**

- su (switch administrative user)
- id (display UID and GID)
- useradd (add user account)
- passwd (change user password)
- userdel / deluser (delete user account)
- usermod (modify user account)
- groupadd (create a user group)
- groupmod (modify group definition)
- groupdel (remove a group)

- gpasswd (administer group passwords and members)

- **Fields of /etc/passwd File:**

1. Username
2. Password (x marker)
3. User ID (UID)
4. Group ID (GID)
5. User Info (GECOS / Full Name)
6. Home Directory
7. Shell

- **Reserved UID Ranges:**

- UID 0: Reserved for root user
- UID 1–99: Predefined system accounts
- UID 100–999: System accounts and groups
- UID 1000+: Regular user accounts

- **Types of Linux Groups:**

1. Primary or login group
2. Secondary or supplementary group

- **Linux Filter Commands:**

- grep (Global Regular Expression Print)
- wc (word, line, character count)
- od (octal dump / format display)
- sort (sort lines of text files)

- **Compress & Decompress Tools & Options:**

- tar options:
 - c: Create an archive
 - x: Extract an archive
 - v: Verbose progress
 - f: Filename of the archive
 - t: View content of the archive
 - j: Filter archive through bzip2
 - z: Filter archive through gzip
 - r: Append or update files
 - W: Verify an archive
 - C: Extract to a different directory
- gzip (compress file to .gz)
- gunzip (decompress .gz file)
- zcat (view compressed files)

- **Linux Utility Commands:**

- find (search files in real-time directory tree)
- locate (search files using pre-indexed database)
- date (display date, time, and timezone)
- cal (display calendar)
- time (measure command execution time)
- df (disk free / report file system disk space usage)
- mount (mount storage device filesystem)
- exit (exit current shell session)
- clear (clear the terminal screen)

- **Linux Networking Commands:**

- ip (manage network interfaces and IP addresses)
- ssh (secure remote command line session)
- mail (send emails from terminal)
- ping (test network connectivity using ICMP)
- host (perform DNS lookups)

- **Link Commands:**

- ln -s (create symbolic/soft link)
- link / ln (create hard link)
- unlink (remove hard link/file entry)